

Business Insights Report

Prepared by: Kiran Kansla

Project Type: Business Analytics • Banking Fraud Detection • Machine Learning

Tools Used: Python • Pandas • Scikit-learn • Matplotlib • NumPy

TABLE OF CONTENTS

1. Executive Summary
2. Business Problem
3. Dataset Overview
4. Fraud Distribution Analysis
5. Customer Transaction Behaviour
6. Risk Driver Analysis
7. Machine Learning Performance
8. Key Business Insights
9. Strategic Recommendations
10. Business Impact
11. Project Conclusion

Executive Summary

Financial fraud remains one of the most significant operational and financial risks for modern banking institutions. With increasing adoption of digital banking services, detecting fraudulent transactions quickly and accurately has become essential to protecting customers and reducing financial losses.

This project analyzed **10,000 simulated banking transactions** using Python and Machine Learning to identify fraud patterns, evaluate transaction behavior, and develop a predictive fraud detection model.

Three machine learning algorithms were developed and compared, with **Random Forest achieving the highest overall performance of 95.25% accuracy**. The analysis also identified several key behavioral indicators—including anomaly score, geographic transaction distance, and device risk score—that were more effective predictors of fraud than transaction amount alone.

The findings demonstrate how data-driven fraud detection can support faster investigations, reduce financial losses, and improve customer security.

Business Problem

Banks process millions of digital transactions every day across mobile applications, internet banking, ATMs, and point-of-sale systems. Traditional rule-based fraud detection methods often fail to identify sophisticated fraud patterns and may generate excessive false alerts.

The objective of this project was to:

- Understand transaction behavior.
- Identify the primary drivers of fraudulent transactions.
- Develop an accurate fraud prediction model.
- Provide actionable recommendations for improving fraud monitoring.

Dataset Overview

The dataset contains **10,000 banking transactions**, each representing a customer transaction with behavioral, transactional, and risk-related information.

Dataset Highlights

- Total Transactions: **10,000**
- Fraudulent Transactions: **1,251**
- Legitimate Transactions: **8,749**
- Fraud Rate: **12.5%**

The dataset includes transaction information such as:

- Transaction Amount
- Payment Channel
- Authentication Method
- Device Risk Score
- AI Anomaly Score

-
- Geographic Distance
 - Transaction Velocity
 - Login Attempts
 - International Transaction Status
 - Suspicious IP Indicator

These variables provide a comprehensive representation of customer behavior and transaction risk.

Fraud Distribution Analysis

The fraud distribution analysis shows that **12.5%** of all transactions were classified as fraudulent, while **87.5%** were legitimate.

Although fraudulent transactions represent a relatively small proportion of the total dataset, they can have a significant financial impact if not detected promptly.

Business Insight

The relatively low fraud rate indicates a class imbalance commonly observed in banking datasets. This highlights the importance of using machine learning models capable of identifying fraudulent transactions without generating excessive false alarms.

Customer Transaction Behaviour

Payment Channel Analysis

The analysis indicates that **Mobile App transactions accounted for the highest transaction volume**, followed by Web Banking, POS terminals, and ATMs.

Fraudulent activity was also highest within the Mobile App channel, primarily because it handled the largest number of overall transactions.

Authentication Analysis

OTP authentication was the most frequently used authentication method across the dataset.

Biometric authentication represented the lowest transaction volume.

Although fraud occurred across all authentication methods, no authentication type completely eliminated fraudulent activity, indicating that authentication alone is insufficient for fraud prevention.

Transaction Amount Analysis

The transaction amount boxplot shows that fraudulent and legitimate transactions have similar distributions.

Business Insight

Transaction amount is not a reliable standalone fraud indicator. Instead, fraud detection systems should prioritize behavioral signals rather than simply flagging high-value transactions.

Risk Driver Analysis

Several behavioral variables were evaluated to determine their influence on fraudulent transactions.

Correlation Analysis

The correlation heatmap revealed that **Anomaly Score has the strongest positive relationship with Fraud Flag (correlation ≈ 0.67)**.

All other variables exhibited relatively weak linear relationships, suggesting that fraud is influenced by a combination of behavioral characteristics rather than a single variable.

Feature Importance

The Random Forest model identified the following variables as the most influential predictors:

1. Anomaly Score
2. Geographic Distance
3. Average Monthly Balance
4. Account Age
5. Transaction Amount
6. Device Risk Score
7. Transaction Velocity Score

Business Insight

The AI-generated anomaly score is the strongest indicator of fraudulent behavior. Transactions with unusually high anomaly scores should receive immediate attention within fraud monitoring systems.

Machine Learning Performance

Three classification algorithms were developed and evaluated.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	95.00%	84.00%	77.60%	80.80%
Decision Tree	94.45%	78.14%	77.20%	77.67%
Random Forest	95.25%	82.70%	78.40%	80.49%

The Random Forest model provided the best overall balance between fraud detection accuracy and predictive performance.

ROC Analysis

- Logistic Regression ROC-AUC: **0.979**
- Random Forest ROC-AUC: **0.975**

Both models demonstrated excellent discrimination between fraudulent and legitimate transactions.

Key Business Insights

The analysis produced several important findings:

- Fraud represented **12.5%** of all transactions.
 - Mobile App transactions generated the largest transaction volume and the highest number of fraud cases.
 - OTP authentication remained the most commonly used authentication method.
 - Transaction amount alone was not sufficient to distinguish fraudulent activity.
 - The anomaly score emerged as the strongest fraud indicator with a correlation of approximately **0.67**.
 - Random Forest achieved the highest overall predictive performance with **95.25% accuracy**.
 - Geographic distance and device risk score were significant contributors to fraud prediction.
 - Behavioral analytics substantially improved fraud detection capability compared with relying solely on transaction value.
-

Strategic Recommendations

Based on the findings, the following recommendations are proposed:

Implement Risk-Based Authentication

Transactions with high anomaly scores or elevated device risk should trigger additional verification methods such as biometric authentication or multi-factor authentication.

Prioritize High-Risk Transactions

Use AI-generated anomaly scores to rank transactions by fraud risk, allowing investigators to focus on the most suspicious cases first.

Strengthen Mobile Banking Security

As Mobile App transactions account for the highest transaction volume and fraud occurrence, enhanced security controls should be implemented for mobile banking channels.

Monitor Geographic Anomalies

Transactions originating from locations significantly different from a customer's historical behavior should be automatically flagged for review.

Continuous Model Retraining

Fraud patterns evolve over time. Regularly retraining machine learning models with new transaction data will maintain predictive performance.

Business Impact

Implementing a machine learning-based fraud detection system provides several operational benefits:

- Faster identification of suspicious transactions.
- Reduced financial losses caused by fraud.
- Improved customer trust through enhanced security.
- Lower investigation costs by prioritizing high-risk alerts.
- Better regulatory compliance through consistent fraud monitoring.
- Improved operational efficiency for fraud investigation teams.
- More accurate allocation of investigative resources.

Overall, integrating machine learning into fraud detection enables financial institutions to shift from reactive investigations to proactive fraud prevention.

Project Conclusion

This project demonstrates the value of combining data analytics and machine learning to address modern banking fraud challenges. By analyzing transaction behavior and comparing multiple predictive models, the project successfully identified the key factors associated with fraudulent activity and developed a highly accurate fraud detection solution.

The Random Forest model achieved **95.25% accuracy**, while the analysis confirmed that behavioral indicators such as anomaly score, geographic distance, device risk score, and transaction velocity provide stronger predictive power than transaction amount alone.

The insights generated through this project can support the development of intelligent fraud monitoring systems capable of reducing financial losses, improving operational efficiency, and strengthening customer trust in digital banking services.